Day 161: The Bias-Variance Tradeoff Revisited, with Real Learning Curves

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 1 | Complete study guide

Watch: Machine Learning Fundamentals: Bias and Variance (StatQuest with Josh Starmer) | Lecture 08 - Bias-Variance Tradeoff (Caltech, Learning From Data)

Script: `day161_bias_variance_learning_curves.py` · every number in this guide is copied by program from `day161_run_output.txt`, the real output of that script.

Week 24 is about judging models honestly, and it starts with the oldest question: is a model wrong because it is too simple, or because it learned its training set's quirks? Both are measured directly on the UCI Adult census data for three models. Logistic regression's training and validation curves meet at a high level (bias: its variance falls from 0.0186 to 0.0005 while its bias term stays near 0.110). One deep tree is almost perfect on its training rows and worse than guessing the base rate on new ones until 16,000 rows (variance up to 0.1138). A first draft of the decomposition made bias appear to rise with tree depth (0.0994 to 0.1056); corrected for using only 12 models, it levels off. Learning curves fitted on small training sets predicted the full-size result to within 0.0040.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

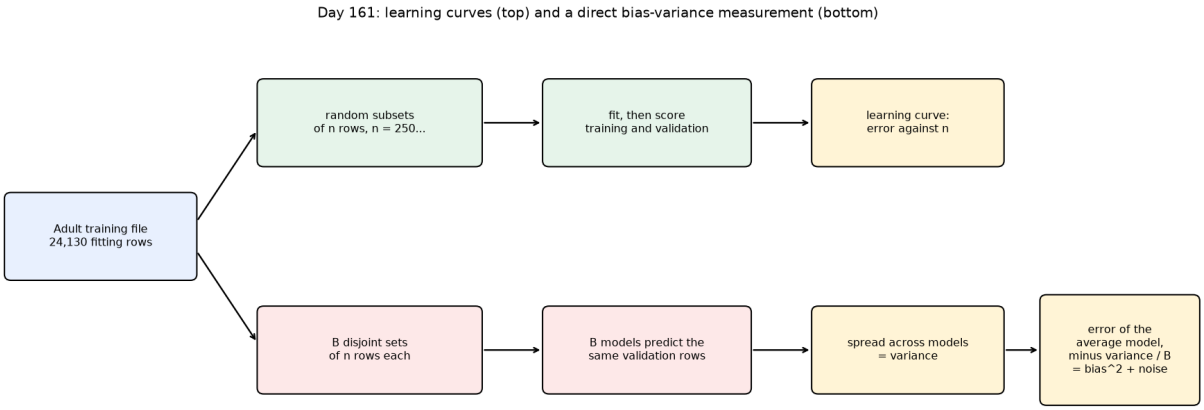
A model's error on new data has three sources. **Bias**: the model is systematically wrong, because it cannot express the pattern (a straight line through a curve). **Variance**: the model changes a lot when trained on a different sample, because it fits that sample's noise. **Noise**: the part nobody can predict (two people with identical records can earn different incomes).

A **learning curve** plots error against the number of training rows, once on the training rows and once on held-out rows. High bias shows as two curves that meet early at a high level; high variance as a wide gap that closes slowly. Today both are also measured directly, by training the same model on many separate samples.

REAL-WORLD SCENARIO

A bank's income model is not accurate enough. Should it buy more data, or build a bigger model? More data cures variance; a bigger model cures bias; the wrong choice wastes money. A learning curve answers the question before the money is spent.

SYSTEM DIAGRAM



Top: learning curves from random subsets. Bottom: the decomposition, from many disjoint training sets predicting the same validation rows.

Week 24 uses one dataset all week: UCI Adult (Kohavi and Becker, 1996), 30,162 people from the 1994 US census after rows with missing values are dropped, with the target *income above $50K* (24.9% of rows). The official training file is split 80/20 into fitting and validation rows; the official test file waits for later in the week. Models come from earlier weeks where possible: Day 109's gradient boosting and regression tree are imported from its exercise file.

Today's part for the week's build

`day161_exercise.py` is today's parts (the data loader and split, the two feature matrices, Newton logistic regression, the Brier score and the decomposition), checked by hand. It is the first part of Day 167's report card.

ACTUAL OUTPUT

```
tree_matrix: sex coded by its rate: F -> 0.0, M -> 1.0 (by hand: 0/2 and 2/2)
onehot_matrix: age (20, 40, 60, 30) standardised -> [-1.183, 0.169, 1.521, -0.507] (by hand: mean
    37.5, spread 14.79); sex columns F, M -> [[1, 0], [0, 1], [0, 1], [1, 0]]
fit_logistic on two groups with 20% and 70% positives: probabilities [0.2, 0.7] in 5 steps (by
    hand: 0.2 and 0.7, so intercept log(0.2/0.8) = -1.3863, got -1.3863)
bias_variance, models (0.2, 0.9) and (0.6, 0.5), targets (0, 1): by hand Brier (0.04 + 0.01 + 0.36
    + 0.25)/4 = 0.165; average model (0.4, 0.7) errs (0.16 + 0.09)/2 = 0.125; spread per row 0.08
    and 0.08 (n - 1 = 1), so variance 0.08 x 1/2 = 0.04 and bias^2 + noise 0.125 - 0.08/2 = 0.085
  bias_variance: Brier 0.165 = 0.085 + 0.040
load_adult: 30162 training-file rows, 15060 test-file rows; split_rows: 24130 fitting, 6032
    validation
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The data, the split, and the models

DEFINITION — start here, no prior knowledge assumed

Trees get one column per feature (12), with each category coded by its share above $50K in the fitting rows. Logistic regression gets 87 columns: the five numbers standardised (capital gains and losses logged first) and one 0/1 column per category.

Logistic regression is fitted by Newton's method: each step uses the curvature (Hessian) of the log loss as well as its slope, and lands on the exact optimum in a few steps. The first draft reused Day 116's Adam optimiser at its own settings instead.

REAL-WORLD SCENARIO

A survey firm predicts which households earn more than $50K from their census answers, to decide who receives a mailing.

MATHEMATICS

$$p = 1 / (1 + e^{-(w_0 + x \cdot w)}); \quad loss = -mean[y \log p + (1-y) \log(1-p)] + (\lambda/2)|w|^2$$

$$Newton: w \leftarrow w - H^{-1} g, \quad g = A^T(p - y)/n + \lambda w, \quad H = A^T \, diag(p(1-p)) \, A / n + \lambda I$$

IMPLEMENTATION

```
def fit_logistic(X, y, lam=LAM, iters=50):
    A = np.c_[np.ones(len(X)), X]; w = np.zeros(A.shape[1])
    R = lam * np.eye(A.shape[1]); R[0, 0] = 0.0
    for it in range(1, iters + 1):
        p = 1.0 / (1.0 + np.exp(-A @ w))
        g = A.T @ (p - y) / len(y) + R @ w
        H = (A * (p * (1 - p))[:, None]).T @ A / len(y) + R
        step = np.linalg.solve(H, g); w = w - step
        if np.abs(step).max() < 1e-9:
            break
    return w, it
```

ACTUAL OUTPUT

```
UCI Adult: 30162 training-file rows and 15060 test-file rows without missing values
share above $50K: training file 0.249, test file 0.246
Week 24's split of the training file: 24130 fitting rows, 6032 validation rows (seed 0)
features: 12 columns for trees, 87 one-hot columns for logistic regression
logistic regression, Newton: 8 steps, largest gradient left 5.2e-17; validation log loss 0.3410, error 0.1592 (0.2 s)
  first draft, Day 116's Adam, 500 steps at rate 0.5: validation log loss 0.4070, error 0.1928
  first draft, Day 116's Adam, 500 steps at rate 0.1: validation log loss 0.3403, error 0.1596
  first draft, Day 116's Adam, 5000 steps at rate 0.1: validation log loss 0.3402, error 0.1601
always predicting the fitting rows' share (0.249): validation Brier 0.1869
Actual output from this lesson's run.
```

What it means

Newton converged in 8 steps with the gradient at 5.2e-17: validation log loss 0.3410, error 0.1592. **First draft:** Day 116's Adam at its own rate of 0.5 stopped at 0.4070; at rate 0.1 it reached 0.3403, the same answer. A rate tuned for one problem did not transfer to another. Always predicting the base rate scores Brier 0.1869: any useful model must beat that.

WHY CHOOSE THIS?

Newton's method needs no learning rate and reaches the exact optimum in a handful of steps for a few hundred columns.

WHY NOT THIS? (TRADEOFFS)

Each step solves a columns × columns system, which becomes slow with many thousands of columns; then gradient methods win.

2. Learning curves

DEFINITION — start here, no prior knowledge assumed

For each training-set size n from 250 up to all 24,130 fitting rows, five random subsets of n rows are drawn (one at the full size), each model is trained, and its Brier score is measured on its own training rows and on the 6,032 validation rows. The **Brier score** is the mean squared error of the predicted probability; unlike log loss it stays finite when a pure tree leaf says 0% or 100%.

Gradient boosting: 200 trees of depth 3 at rate 0.1. The deep tree is grown until every leaf is pure.

REAL-WORLD SCENARIO

A student's practice-test score against their real-exam score, as they study more past papers.

MATHEMATICS

$$Brier = (1/n)\,\Sigma\,(p_i - y_i)^2; \quad \text{always predicting the base rate } \pi \text{ scores } \pi(1-\pi)$$

ACTUAL OUTPUT

```
Brier score (mean squared error of the probability), mean of 5 random subsets (1 at the full 24130); training / validation
  logistic regression (1 s)
    250: 0.087 / 0.130; 500: 0.091 / 0.119; 1000: 0.104 / 0.114; 2000: 0.106 / 0.113; 4000: 0.108 / 0.111; 8000: 0.107 /
        0.111; 16000: 0.108 / 0.110; 24130: 0.108 / 0.110
  gradient boosting (55 s)
    250: 0.007 / 0.142; 500: 0.029 / 0.122; 1000: 0.049 / 0.108; 2000: 0.066 / 0.102; 4000: 0.076 / 0.097; 8000: 0.082 /
        0.095; 16000: 0.085 / 0.093; 24130: 0.087 / 0.092
  one deep tree (8 s)
    250: 0.000 / 0.229; 500: 0.001 / 0.217; 1000: 0.001 / 0.208; 2000: 0.003 / 0.200; 4000: 0.004 / 0.195; 8000: 0.007 /
        0.188; 16000: 0.010 / 0.184; 24130: 0.013 / 0.182
validation error rate at the full size: logistic regression 0.1592, gradient boosting 0.1338, one deep tree 0.1897
Actual output from this lesson's run.
```

What it means

Logistic regression's curves meet: training 0.108 and validation 0.110 at the full size, with little gain after 2,000 rows. That is the signature of **bias**: more data will not help, a more flexible model might. Boosting's gap is wider and still closing (0.087 / 0.092): it is better, and a little more data would still help. The deep tree is the **variance** case: training Brier 0.000 at 250 rows, validation 0.229, still worse than the base rate at 8,000 rows (0.188) and better only from 16,000.

As error rates at the full size: boosting 0.1338, logistic regression 0.1592, the deep tree 0.1897. A learning curve shows the shape that a single number hides.

WHY CHOOSE THIS?

Cheap to draw, and it answers the practical question: buy data, or change the model?

WHY NOT THIS? (TRADEOFFS)

Needs many refits; the curve at small n depends on the random subsets (five here), so read the trend, not one point.

3. Bias and variance, measured directly

DEFINITION — start here, no prior knowledge assumed

Train the same model on B training sets of n rows that share no rows, and let all B models predict the same validation rows. For each row, the spread of the B predictions is the **variance**, and the error of their average is **bias² + noise** (noise cannot be separated without knowing each person's true probability).

The average of only B models still carries some of their spread (variance / B), so it is subtracted. B is as many disjoint sets as fit in 24,130 rows, up to 20; at 8,000 rows only 3 fit.

REAL-WORLD SCENARIO

Twenty survey teams each interview a different random 1,000 households and fit the same model. Where their predictions disagree is variance; where they agree but are wrong is bias.

MATHEMATICS

$$Brier = mean_{models,\ rows}(p - y)^2 = [mean_{rows}(p_{avg} - y)^2 - s^2/B] + s^2(B-1)/B$$

p_{avg} = *average of the B predictions;* s^2 = *their variance (dividing by B – 1), averaged over rows*

IMPLEMENTATION

```
def bias_variance(P, y):
    B = len(P); var = P.var(0, ddof=1)
    bias_noise = np.mean((P.mean(0) - y) ** 2) - var.mean() / B
    return float(np.mean((P - y) ** 2)), float(bias_noise), float(var.mean() * (B - 1) / B)
```

ACTUAL OUTPUT

```
Brier on validation = (bias^2 + noise) + variance; B = disjoint training sets of n rows
  logistic regression:
    n 250 (B 20): 0.1299 = 0.1104 + 0.0186; n 500 (B 20): 0.1201 = 0.1102 + 0.0094; n 1000 (B 20): 0.1149 = 0.1099 + 0.0048;
        n 2000 (B 12): 0.1126 = 0.1099 + 0.0025; n 4000 (B 6): 0.1112 = 0.1099 + 0.0010; n 8000 (B 3): 0.1106 = 0.1099 +
        0.0005
  gradient boosting:
    n 250 (B 20): 0.1416 = 0.1002 + 0.0394; n 500 (B 20): 0.1231 = 0.0975 + 0.0243; n 1000 (B 20): 0.1098 = 0.0949 + 0.0141;
        n 2000 (B 12): 0.1026 = 0.0937 + 0.0082; n 4000 (B 6): 0.0977 = 0.0930 + 0.0039; n 8000 (B 3): 0.0949 = 0.0922 +
        0.0018
  one deep tree:
    n 250 (B 20): 0.2207 = 0.1009 + 0.1138; n 500 (B 20): 0.2152 = 0.1012 + 0.1083; n 1000 (B 20): 0.2018 = 0.0949 + 0.1015;
        n 2000 (B 12): 0.2029 = 0.0975 + 0.0966; n 4000 (B 6): 0.1982 = 0.0952 + 0.0858; n 8000 (B 3): 0.1862 = 0.0880 +
        0.0655
Actual output from this lesson's run.
```

What it means

Logistic regression's bias term barely moves (0.1104 to 0.1099) while its variance falls from 0.0186 to 0.0005. Boosting's variance falls the same way (0.0394 to 0.0018) and its bias term also falls a little (0.1002 to 0.0922). The deep tree's bias term is the lowest of the three at 0.0880, but its variance is enormous (0.1138 at 250 rows, still 0.0655 at 8,000).

The deep tree's bias estimates jump about (0.0949, then 0.0975) because subtracting variance / B is noisy when variance is large and B is small. A likely reason its bias is lowest, not tested here: averaging many pure trees is what a random forest does, and its average is a flexible, nearly unbiased model.

WHY CHOOSE THIS?

Measures the two causes separately instead of guessing them from a gap.

WHY NOT THIS? (TRADEOFFS)

Needs many disjoint training sets, which runs out at large n (B = 3 at 8,000 rows); noise stays mixed into the bias term.

4. A complexity sweep, and a first-draft error

DEFINITION — start here, no prior knowledge assumed

One tree, depth 1 to 20, on the same 12 disjoint sets of 2,000 rows. The first draft used the error of the average model as the bias term, without subtracting variance / B.

REAL-WORLD SCENARIO

Tuning how detailed a map is: too coarse misses streets, too fine records every parked car that will have moved tomorrow.

ACTUAL OUTPUT

```
one tree, 12 disjoint training sets of 2000 rows; validation Brier = (bias^2 + noise) + variance
  depth  1: 0.1494 = 0.1492 + 0.0002   first draft's bias^2 + noise: 0.1492
  depth  2: 0.1285 = 0.1236 + 0.0045   first draft's bias^2 + noise: 0.1240
  depth  3: 0.1187 = 0.1117 + 0.0064   first draft's bias^2 + noise: 0.1123
  depth  4: 0.1155 = 0.1065 + 0.0082   first draft's bias^2 + noise: 0.1073
  depth  5: 0.1144 = 0.1021 + 0.0113   first draft's bias^2 + noise: 0.1031
  depth  6: 0.1149 = 0.0997 + 0.0139   first draft's bias^2 + noise: 0.1010
  depth  8: 0.1232 = 0.0972 + 0.0239   first draft's bias^2 + noise: 0.0994
  depth 10: 0.1393 = 0.0964 + 0.0393   first draft's bias^2 + noise: 0.1000
  depth 12: 0.1571 = 0.0966 + 0.0555   first draft's bias^2 + noise: 0.1016
  depth 14: 0.1720 = 0.0964 + 0.0693   first draft's bias^2 + noise: 0.1027
  depth 16: 0.1845 = 0.0966 + 0.0806   first draft's bias^2 + noise: 0.1040
  depth 20: 0.1973 = 0.0973 + 0.0917   first draft's bias^2 + noise: 0.1056
best depth 5 (Brier 0.1144); from depth 1 to 20 the corrected bias^2 + noise falls 0.1492 -> 0.0973, the first draft's
    0.1492 -> 0.1056
Actual output from this lesson's run.
```

What it means

The classic U: validation Brier falls to its best at depth 5 (0.1144) and then climbs to 0.1973 at depth 20. Bias² + noise falls from 0.1492 and levels off near 0.097; variance rises from 0.0002 to 0.0917.

First draft: the uncorrected bias term bottomed out at depth 8 (0.0994) and rose again to 0.1056, as if deeper trees became more biased. That rise was variance: the average of 12 models still carries a twelfth of it. Corrected, the bias term is flat (0.0964 to 0.0973).

WHY CHOOSE THIS?

The sweep finds the depth where the two causes balance, and shows which one dominates on each side.

WHY NOT THIS? (TRADEOFFS)

With few models every estimate is noisy; finite-sample corrections are easy to forget, as the first draft shows.

5. Can a learning curve be extrapolated?

DEFINITION — start here, no prior knowledge assumed

Fit validation Brier = a + b n^{-c} to the curve's small sizes only (up to 2,000 or 4,000 rows), then predict the full 24,130 and compare with what actually happened. *a* is the error the model approaches with unlimited data.

REAL-WORLD SCENARIO

Deciding whether to pay for 20,000 more labelled examples, using a pilot study of 4,000.

MATHEMATICS

$$error(n) \approx a + b\, n^{-c};\quad \text{for each } c \text{ on a grid, } a \text{ and } b \text{ by least squares; keep the } c \text{ with the smallest squared error}$$

ACTUAL OUTPUT

```
validation Brier fitted as a + b n^(-c) on the small sizes only, then predicted at the full size
fitted on sizes up to 2000:
  logistic regression: a 0.1110, b 17.540, c 1.24; predicted at 24130: 0.1111, actual 0.1102 (error +0.0009)
    gradient boosting: a 0.0895, b 2.664, c 0.71; predicted at 24130: 0.0916, actual 0.0918 (error -0.0003)
        one deep tree: a 0.1708, b 0.361, c 0.33; predicted at 24130: 0.1837, actual 0.1825 (error +0.0012)
fitted on sizes up to 4000:
  logistic regression: a 0.1104, b 11.307, c 1.15; predicted at 24130: 0.1105, actual 0.1102 (error +0.0003)
    gradient boosting: a 0.0897, b 2.730, c 0.72; predicted at 24130: 0.0917, actual 0.0918 (error -0.0001)
        one deep tree: a 0.1781, b 0.453, c 0.40; predicted at 24130: 0.1865, actual 0.1825 (error +0.0040)
Actual output from this lesson's run.
```

What it means

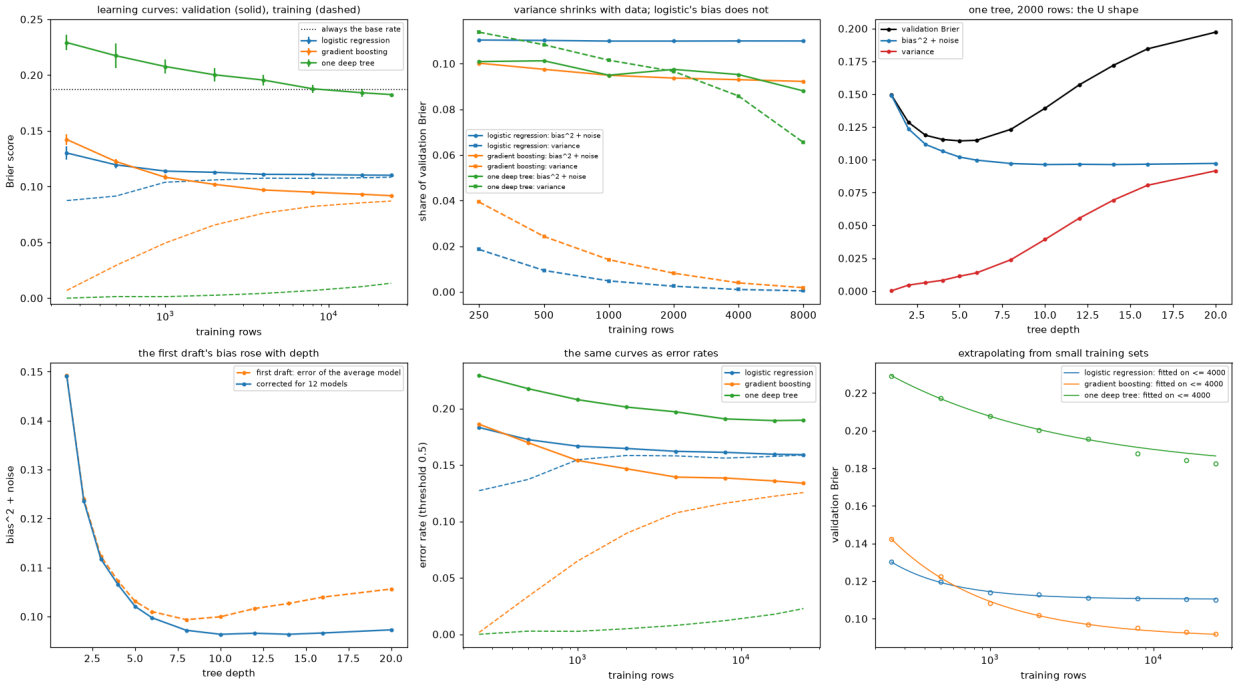
From 4,000 rows, the predictions at 24,130 were off by +0.0003 (logistic regression), -0.0001 (boosting) and +0.0040 (the deep tree, too pessimistic). Fitting on fewer sizes (up to 2,000) did better for the tree (+0.0012) and slightly worse for logistic regression (+0.0009), so the curves are only roughly power laws. For logistic regression the fitted floor a is already reached: no data will improve it.

WHY CHOOSE THIS?

A cheap forecast of what more data is worth, from a small pilot.

WHY NOT THIS? (TRADEOFFS)

An extrapolation, not a measurement: one fit was off by 0.004, and the shape can change outside the measured range.



Top: learning curves (solid validation with the spread over five subsets, dashed training); bias² + noise and variance by training size; the depth sweep's U. Bottom: the first draft's bias against the corrected one; the same learning curves as error rates; power-law fits on up to 4,000 rows against the actual points.

Interview preparation — technical questions

Q: How do you tell high bias from high variance with a learning curve?

A: High bias: training and validation error meet early at a high level (logistic regression here, 0.108 and 0.110). High variance: a large gap that closes slowly (the deep tree, near 0 against 0.18 or more).

Q: What would you do in each case?

A: Bias: a more flexible model or better features. Variance: more data, regularisation, a simpler model, or averaging many models.

Q: How can bias and variance be measured, not guessed?

A: Train on many disjoint samples, predict the same held-out rows; the spread is variance, the error of the average (minus variance / B) is bias² + noise.

Q: Why the Brier score instead of log loss here?

A: A pure tree leaf predicts exactly 0 or 1, so one wrong row makes log loss infinite; the Brier score stays bounded and splits exactly into bias and variance.

Q: Why did the optimiser matter for logistic regression?

A: Day 116's Adam at rate 0.5 stopped at log loss 0.4070; Newton found the optimum 0.3410 in 8 steps. Check convergence before comparing models.

Interview preparation — theoretical questions

Q: Is the bias-variance tradeoff always a U?

A: For one tree's depth here, yes. Averaging methods (forests, boosting) and very large networks can reduce variance without raising bias, so the U is not a law.

Q: Why is noise mixed into the bias term?

A: Separating them needs each person's true probability of earning above $50K, which no dataset records.

Q: Why does a finite number of models bias the decomposition?

A: The average of B predictions still varies by variance / B; without subtracting it, bias appears to rise with variance.

Q: Can a learning curve predict the value of more data?

A: Roughly: a power-law fit on up to 4,000 rows predicted the full size within 0.004 here, but it is an extrapolation.

Key takeaway and what's next

> **KEY TAKEAWAY**
> Bias and variance are not abstractions: both can be measured, and learning curves show which one limits a model. Logistic regression is limited by bias, a deep tree by variance, and boosting balances them best on this data. Two first drafts went wrong in instructive ways: a borrowed optimiser setting, and a decomposition that forgot how few models it averaged.

- Full-size validation Brier: boosting 0.092, logistic regression 0.110, deep tree 0.182; base rate 0.1869.
- Logistic regression's variance 0.0186 → 0.0005; its bias term flat near 0.110.
- Tree depth: best at 5; the uncorrected bias term rose to 0.1056, the corrected one stayed near 0.097.
- Power-law extrapolation from 4,000 rows: within 0.0040 of the actual full-size Brier.

Suggested watch order, if starting from zero

- 1. Machine Learning Fundamentals: Bias and Variance (StatQuest with Josh Starmer) — the two ideas with simple pictures.
- 2. Lecture 08 - Bias-Variance Tradeoff (Caltech, Learning From Data) — the full derivation, with learning curves, from Caltech's Learning From Data course.

Day 162 preview

- **Hypothesis testing for ML.** A real randomised experiment (Hillstrom's email campaign, 64,000 customers) is analysed as an A/B test: two-proportion tests, permutation tests, an A/A check, power, and McNemar's test for comparing two models on the same rows.